DSBDA Practical Viva Guide

Ayan Sheikh

SPPU TE Computer Engineering

Experiment 1 — Data Wrangling I

Q1. What is Data Wrangling?

Data wrangling means cleaning and converting raw data into a usable format for analysis. It includes handling missing values, normalization, encoding, and datatype conversion.

Q2. What is preprocessing?

Preprocessing means preparing data before analysis or machine learning. It includes removing null values, scaling data, and converting text into numbers.

Q3. What is a null value?

A null value is a missing or empty value in a dataset. In pandas, NaN represents missing data.

Q4. How do you find null values?

We can find null values using df.isnull().sum(). It shows missing values in each column.

Q5. How do you handle missing values?

Missing values can be removed using dropna() or filled using fillna(). Mean is used for numerical data and mode for categorical data.

Q6. What is normalization?

Normalization scales data into a smaller range, usually between 0 and 1, so all values become comparable.

Experiment 2 — Data Wrangling II

Q1. What is an outlier?

An outlier is a value that is very far from normal data values. Example: Age = 500.

Q2. How do you detect outliers?

Outliers can be detected using boxplots, IQR method, or Z-score method.

Q3. What is IQR?

IQR stands for Interquartile Range. It is calculated using Q3 - Q1.

Q4. How do you handle outliers?

Outliers can be removed, replaced, or capped using methods like clip().

Q5. What is skewness?

Skewness means asymmetry in data distribution. Right skew has a tail on the right side.

Experiment 3 — Descriptive Statistics

Q1. What is mean?

Mean is the average value of all observations in the dataset.

Q2. What is median?

Median is the middle value after sorting the data.

Q3. What is mode?

Mode is the value that appears most frequently in the dataset.

Q4. What is standard deviation?

Standard deviation measures how much data values are spread from the mean.

Q5. What is variance?

Variance is the square of standard deviation.

Q6. What is percentile?

Percentile shows the position of data values. Example: 75th percentile means 75% values are below it.

Mean	Median
Affected by outliers	Less affected by outliers
Average value	Middle value

Experiment 8/9/10 — Visualization

Q1. What is data visualization?

Data visualization means displaying data graphically using charts and plots.

Q2. Why visualization is important?

Visualization helps understand patterns, trends, and outliers easily.

Q3. What is histogram?

Histogram shows the frequency distribution of numerical data.

Q4. What is countplot?

Countplot displays the frequency of categorical values.

Q5. What is pairplot?

Pairplot shows relationships between multiple variables in a dataset.

Logistic Regression

Q1. What is Logistic Regression?

Logistic Regression is a supervised machine learning algorithm used for classification problems.

Q2. What is classification?

Classification means predicting categories or classes like Yes/No or True/False.

Q3. What is train_test_split()?

train_test_split() divides the dataset into training data and testing data.

Q4. What is confusion matrix?

Confusion matrix is a table showing correct and incorrect predictions.

Q5. What is accuracy?

Accuracy is the percentage of correct predictions made by the model.

Naive Bayes

Q1. What is Naive Bayes?

Naive Bayes is a probability-based classification algorithm.

Q2. Why called Naive?

It is called Naive because it assumes all features are independent.

Q3. Applications of Naive Bayes?

Naive Bayes is used in spam detection, sentiment analysis, and text classification.

Q4. Why is Naive Bayes fast?

Naive Bayes is fast because it uses simple probability calculations.